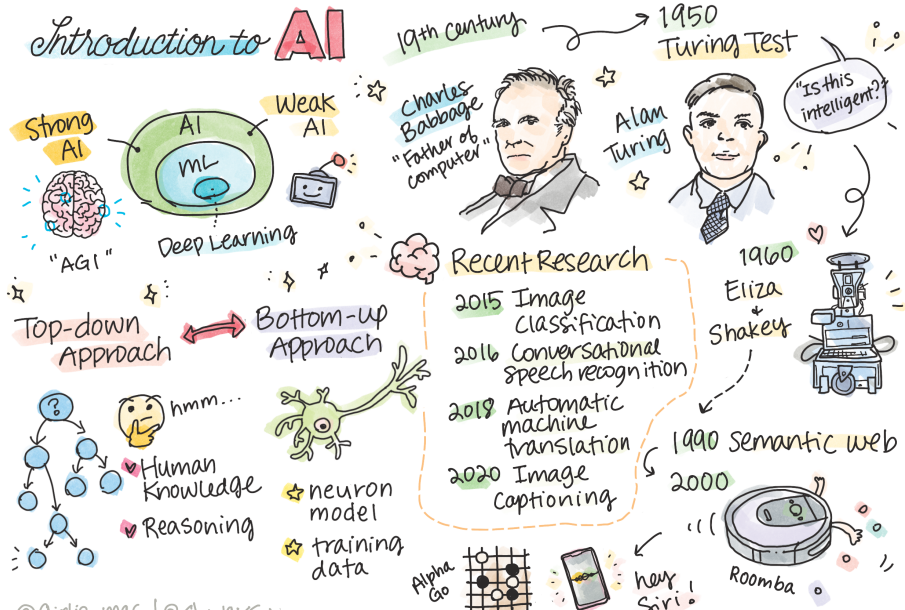


Introduction to AI

AI for Beginners Course

Introduction to AI



What is Artificial Intelligence?

Artificial Intelligence is an exciting scientific discipline that studies how we can make computers exhibit intelligent behavior.

- Computers operate on algorithms - well-defined procedures
- Some tasks cannot be explicitly programmed
- Example: Determining age from a photograph
We cannot explicitly explain how we determine age, nor can we program a computer to do it traditionally.

Weak AI vs. Strong AI

Weak AI (Narrow AI)

- Designed for specific tasks
- Examples: Siri, Alexa, recommendation algorithms
- Highly specialized, lacks general intelligence

Strong AI (AGI)

- Human-level intelligence and understanding
- Can perform any intellectual task
- Currently theoretical - not yet achieved

Defining Intelligence

- No clear universal definition exists
- Connected to abstract thinking and self-awareness
- Difficult to measure objectively

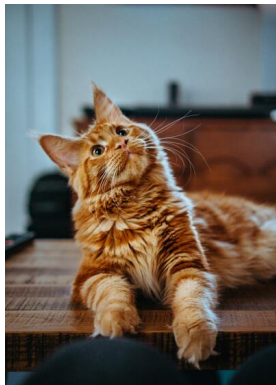


Figure 2: Photo of a Cat

Question: Is a cat intelligent? Different people give different an-

The Turing Test

Proposed by Alan Turing as a test for machine intelligence:

- Compares system to human intelligence
- Human interrogator conducts text-based dialogue
- If interrogator cannot distinguish machine from human, system is considered intelligent

Eugene Goostman (2014):

- Convinced 30% of judges it was human
- Posed as 13-year-old Ukrainian boy
- Demonstrated cleverness, not true intelligence

Two Main Approaches to AI

Top-down Approach (Symbolic Reasoning)

- Models human reasoning processes
- Extracts knowledge from experts
- Represents knowledge in computer-readable form
- Uses explicit rules and logic

Bottom-up Approach (Neural Networks)

- Models structure of human brain
- Uses interconnected neurons
- Learns from training data
- Similar to how babies learn

Emergent/Synergetic Approach:

- Complex behavior from simple agent interactions
- Intelligence emerges from reactive behaviors

Evolutionary Approach:

- Optimization based on evolution principles
- Genetic algorithms

Brief History of AI

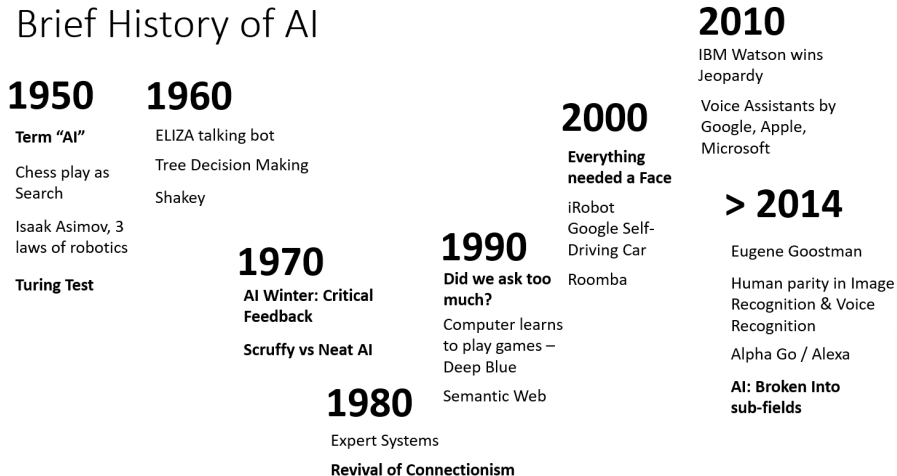


Figure 3: Brief History of AI

Brief History of AI

- **1950s-1960s:** Birth of AI, symbolic reasoning dominates
- **1970s:** AI Winter - expert systems too expensive
- **2010s:** Neural networks rise with big data
- **Today:** AI mostly synonymous with neural networks

Evolution: Chess Programs

Early Programs:

- Search-based algorithms
- Alpha-beta pruning

Middle Era:

- Case-based reasoning
- Learning from human matches

Modern Programs:

- Neural networks
- Reinforcement learning
- Self-play and learning from mistakes

Evolution: Conversational AI

Turing Test Evolution

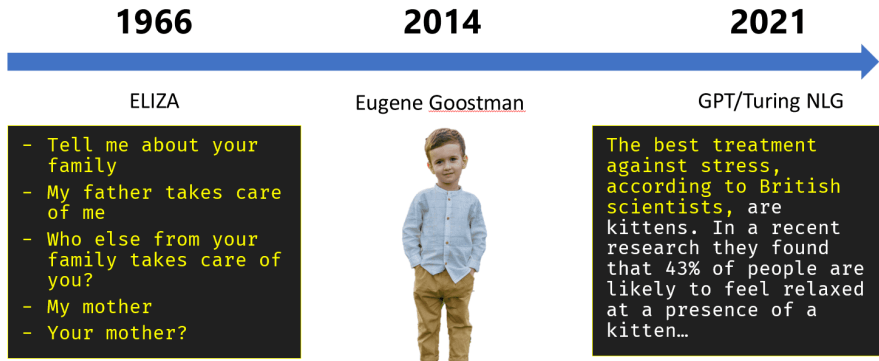
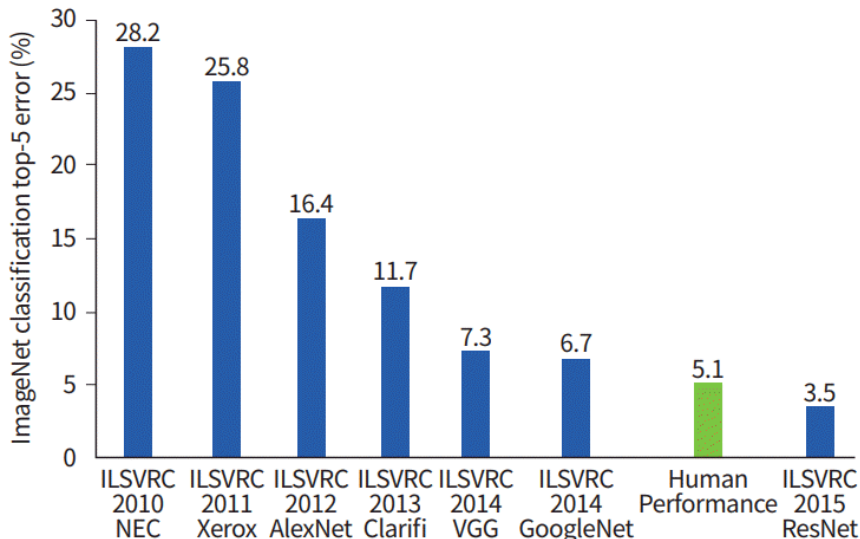


Figure 4: The Turing test's evolution

Early: ELIZA - simple grammatical rules

Recent AI Research (2012-2020)

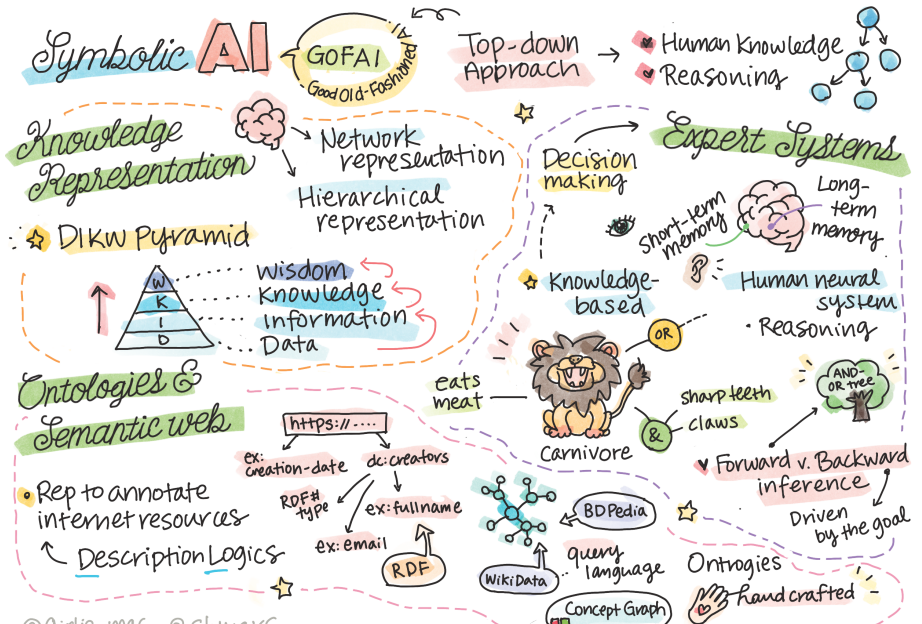


Human Parity Achievements

Year	Achievement
2015	Image Classification
2016	Conversational Speech Recognition
2018	Automatic Machine Translation
2020	Image Captioning

Recent Success: Large language models (BERT, GPT-3)

Knowledge Representation and Expert Systems



Key Takeaways

- ① AI aims to make computers exhibit intelligent behavior
- ② Two main approaches: symbolic reasoning and neural networks
- ③ Neural networks currently dominate AI research
- ④ Massive datasets enabled recent breakthroughs
- ⑤ AI has achieved human parity in multiple domains
- ⑥ Large language models represent the cutting edge

Challenge:

- Research where AI is most effectively used
- Investigate how these systems were built

Review:

- Explore the history of AI and ML
- Research cultural context of AI evolution
- Complete the Game Jam assignment

- Pre-lecture quiz: <https://ff-quizzes.netlify.app/en/ai/quiz/1>
- Post-lecture quiz: <https://ff-quizzes.netlify.app/en/ai/quiz/2>
- ML for Beginners: <http://aka.ms/ml-beginners>
- Assignment: Game Jam